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Population viability risk management (PVRM) for in situ management of endangered tree species—A case study on a *Taxus baccata* L. population

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Abstract

Population viability risk management (PVRM) provides a framework for explicitly including qualitative information about the possible outcomes of a management decision with regard to the viability of an endangered population in conservation management. Multi-criteria decision-making (MCDM) techniques enables managers to select the most preferred choice of action in a context where several criteria apply simultaneously. In that context a combined approach of the PVRM concept and a MCDM technique is presented for the development, evaluation and finally ranking of the in situ conservation strategies. We discuss the concept based on a case study for the maintenance of a gene conservation forest of an English yew population (*Taxus baccata* L.) in Styria, Austria. As part of the PVRM the analytic hierarchy process (AHP) is used to evaluate six conservation strategies with regard to the viability of the yew population. The viability of the population is evaluated based on the results of an analysis of the current environmental, social and economical state and a characterization of the ecological parameters of its population. The most significant risk factors (illegal cutting, browsing by game, tree competition, light availability and genetic sustainability) are structured and prioritised according to their impact on the viability of the yew population applying the AHP. Effects of the six conservation strategies on the viability of the yew population are determined through a qualitative assessment of the probability of a decrease of the population along with four different environmental scenarios. In this context strategy IV combining selective thinning, protection measures, game control with public relation activities seems to be the most effective alternative. The benefits of the combined approach of the PVRM concept with the AHP for the rational analysis of conservation strategies for this endangered tree species are discussed.

Keywords: English yew (*Taxus baccata* L.); Analytic hierarchy process (AHP); Forest management; Nature conservation; Endangered species

1. Introduction

In conservation management the assessment of the viability of endangered populations is a major concern for planning and decision-making. Population viability is related to the probability of a continued existence of a geographically well-distributed population over a specified time period, whereas population viability analysis (PVA) is the formal process by which the likelihood of persistence of the population, metapopulation or species over a specified period of time is estimated (Bustamante, 1996; Marcot and Murphy, 1996). According to Malcolm and Hunter (2002) population viability analysis includes a systematic approach to understand the process that makes a population vulnerable to extinction. PVA is neither a monolithic concept nor a cookbook procedure. It was originally introduced by Mark L. Shaffer in 1978 for the evaluation of extinction probabilities of grizzly bears (*Ursus arctos*) in the USA. Applications dealing with risk assessment, hybridisation and management of critically endangered animals, vaccination against infectious diseases, timing and intensity of pest control, frequency and cause of mouse, rat and stoat irruptions in a forest ecosystem, or survival prospects of populations under threat of predation are described (compare Keedwell, 2004). At present population viability analyses for plants are scarce (Pfab and Witkowski, 2000; Menges, 2000). The models and methods vary both in terms of complexity of

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the underlying models and the quantity of data needed to parameterise them (Burgman et al., 1993; Morris et al., 1999; Bessinger, 2002). One of the most limiting factors of PVA is the need for a large quantity of field data (Akcakaya and Sjögren-Gulve, 2000; Ellner et al., 2002). Therefore many of the models used in population viability studies rely on monitoring data from a short period of time which might lead to the fact that not all dynamics of a species population are captured (Fiedler et al., 1998; Menges, 2000). In qualitative assessments of population viability a mix of professional judgment and empirical evidence is used to pose working hypotheses concerning population responses to proposed management actions. The qualitative information is often not directly linked to the duration of study and amount of data needed. Examples of qualitative population viability assessments include the evaluations of the persistence of fungi, lichen, bryophyte, non vascular and vascular plants, invertebrates, fish and wild life species and species groups in recent region-wide conservation planning exercises (Thomas et al., 1993; FEMAT, 1993).

The host of factors that potentiality influences the viability of species, and how those factors are influenced by management activities, thus are case-specific (Marcot and Murphy, 1996). In the context of conservation management, decisions about an appropriate management strategy to maintain a threatened species have to be taken, but most of the population viability studies do not include the management aspect intensively (compare the review given in Menges, 2000). Because resource managers often make decisions with less than complete information, assumptions founded on concepts of conservation management are often substituted for empirical data as a basis for action (Schulte et al., 2006). However, measuring survival rates of management activities for plant conservation in combination with a qualitative population viability analysis is difficult (Menges, 2000).

Decision analysis is a tool originally developed for guiding business decisions under uncertainty (Beha and Vaupel, 1982) but can be used for the management of endangered species as well (Maguire, 1986). It offers a framework where political, financial, and scientific information can be analysed. Ecological theory, objective data, subjective judgements, qualitative information and financial aspects are used to support the management process (Thibodeau, 1983). Multi-criteria decision-making (MCDM) techniques enables resource managers to select the most preferred choice of action in a context where several criteria apply simultaneously. In a rational decision making environment, the most preferred choice is generally bounded by the management objectives, and the constraints that limit the choices and the achievement of the objectives (Mendoza and Prabhu, 2000). The analytical hierarchy process (AHP) is a MCDM method which allows the pairwise comparison of management alternatives with respect to single decision criteria based on a ratio scale (Saaty, 1995). Mendoza and Sprouse (1989) are pioneers for using this technique for the analysis of a forest management decision problems and nowadays AHP is applied in a wide array of decision problems related to multi-objective forest management (Mendoza and Prabhu, 2000; Kangas and Kangas, 2002; Wolfslehner et al., 2004) and conservation management (Kangas and Kuusipalo, 1993; Zadnik, 2006; Zhu et al., 2001; Abiyu et al., 2006). However, MCDM techniques are not often used in relation to population viability studies to allow a rationale, transparent and comprehensive planning and decision-making processes.

Marcot and Murphy (1996) introduced the concept of population viability risk management (PVRM), which provides a framework for explicitly including qualitative information about the possible outcomes of a management decision with regard to the viability of a population. Actually the PVRM concept is familiar to the conservation biologist for the management of endangered wildlife populations but recently this method was applied in the conservation of endangered trees species (Menges, 1990; Vacik et al., 2001). Although there are many uncertainties that might caution against believing specific projections of population futures, comparative approaches can distinguish between alternative management strategies in a way that is more comprehensive than considering single life history responses (Menges, 2000).

The objective of this contribution is to present the use of the analytical hierarchy process (AHP) as part of a PVRM framework for endangered tree species. The PVRM approach allows the evaluation of different conservation management strategies in order to maintain species at risk, and to identify courses of action, which improve the viability of the population. We will draw our example of the proposed approach for the management of an endangered English yew (*Taxus baccata* L.) population in Austria. The viability for the English yew population will be assessed based on the results of an analysis of the current environmental, social and economical state. Sensitivity analysis will be used to select a compromise solution out of six in situ conservation strategies.

2. PVRM

There is no universal guideline or single method for conducting PVRM, as all factors potentially influencing the viability of a population are case-specific. However, Marcot and Murphy (1996) proposed a general nine-step approach to guide conservation managers' for selecting conservation strategies. We adapted this general procedure for the management of endangered tree species and linked it to the formal steps of decision analysis as follows.

2.1. Identifying the species at risk and relevant regulations

A preliminary step in conservation planning is the selection of the target species and species relevant regulations and laws. Information about small numbers of species, limited geographic distribution, downward populations trends and sensitivity to human pressures are used to identify the species at risk. Geographic approach to protection (GAP) analysis (Scott et al., 1993) or red and blue lists of endangered species can be used to support this process. This step includes the definition of the range of an acceptable viable condition to meet the policy regulations or legal mandates, and identification of the key agencies or institution that are responsible to carry out the work

Table 1
General importance of ecophysiological characteristics of English yew with regard to risk susceptibility and viability of a population

Ecophysiological characteristics of yew	Comments	Influence on disposition
Slow growth	The slow growth of yew reduces the capabilities to compete with other neighbouring tree species for resources which suppress the overall growth and development (Thomas and Polwart, 2003)	↑↑↑
Vegetative regeneration capacity	A high vegetative regeneration-capacity helps yew to survive after severe damages of trunk and crown and increases the overall chances for natural recruitment as well (Suszka, 1978; Dhar et al., 2006)	↓↓
Hard resistant wood	Yew is slight susceptibility against wood rottenness after stem damages	↓
Weak needles	The needles of yew are intolerant to severe and prolonged frost (Skorupski and Luxton, 1998) and icy wind (Bugala, 1978) and do not protect against high transpiration rates (Leuthold, 1980; Zoller, 1981)	↑↑
Shade tolerance of juvenile plants	Seedlings can survive in low light intensity up to few years after germination, but light is an important factor for its growth and survival in the years to follow (Krol, 1978; Boratynski et al., 1997; Thomas and Polwart, 2003; Iszkulo et al., 2005; Iszkulo and Boratynski, 2006)	↓
Shade tolerance of adult plants	An adult yew can survive in unfavourable light conditions for a long time (Lilpop, 1931; Krol, 1978; Brzeziecki and Kienast, 1994)	↓↓
Resistance to fire and abiotic damages	A high resistance to fire and the high vegetative regeneration-capacity allows yew to survive intensive abiotic damages (Gilman and Watson, 1994)	↓
High drought tolerance	The high drought tolerance allows yew to overcome severe shortcomings in water availability (Gilman and Watson, 1994; Thomas and Polwart, 2003)	↓
Dioecious sexual system	The major advantage of a dioecious sexual system is the reduction of the inbreeding depression (Darwin, 1876). High levels of heterozygoty might be found even in small populations. Disadvantages are the loss of fitness via one or other sex function and the lack of mobility which can lead to extinction of small populations on the other hand (Charlesworth, 2001)	↑
Susceptibility to diseases, pest and biotic damages	Diseases are not a major concern although yew is notably susceptible to *Phytopthera* sp. root diseases (Strouts, 1993) and ramorum dieback (*P. ramorum*) (Lane et al., 2004). Thomas and Polwart, 2003 noted that big bud mite (*Cecidophyopsis psilaspis* Nalepa: Eriophyidae) considered as a serious pest of yew in northern and central Europe. However, *Taxus* mealybug, black vine weevil, *Taxus* scale and can cause some damages (Gilman and Watson, 1994)	↑
Wide physiological amplitude	Yew has a wide physiological amplitude which allows yew to spread out on a wide range of sites (Thomas and Polwart, 2003)	↓↓
Susceptibility to browsing and grazing	Kelly (1981), Haeggström (1990), Mysterud and Østbye (2004) and Dhar et al. (2006) reported that yew is very susceptible to browsing and Watt (1926) pointed out that grazing can drastically affect the net growth rates which leads to strong negative effects on recruitment and adult survival in deer populated areas	↑↑↑

(↑) Increase disposition (risk of extinction); (↓) decrease disposition (risk of extinction).

and coordinate their activities. It might include a formal definition of a minimal viable population (MVP); a contingent upon decisions of tolerable risk of extinction and a defined time period for a give population at a certain site (Shaffer, 1981). Many studies try to avoid the calculation of a MVP, as the concept does not include uncertainties regarding the data used and other model assumptions (Menges, 2000).

2.2. *Description of the ecological conditions of the target species and their environment*

In this phase detailed information regarding the general ecophysiological characteristics of the target species and the specific environmental conditions at a given site is mandatory to determine the viability of the population. Literature studies, expert know how and experiences from long term monitoring plots are used to describe the ecophysiological characteristics (e.g. growth potential, light requirements, regeneration strategy, sexual system, susceptibility to damages and infestations) of the tree species (for detailed information see Table 1). Information regarding the history of forest management, the population structure, the socio-economic conditions and possible reasons for viability concerns or decline of the populations have to be identified. The concept of ‘‘risk of extinction’’ is often used to define the targets for recovery by estimating the loss of reproductive capacity or loss of genetic diversity. Field inventories and investigations should help to describe the population size (number of species ha^{-1}), sex ratio (male/female), vitality (health condition of the tree by means of crown length, defoliation and damage), structure (regarding tree height, diameter distribution), human interventions (e.g. illegal logging) and public awareness (see also Morrison et al., 1992). With this information a comprehensive qualitative assessment according to different levels of the population can be undertaken (see FEMAT, 1993; Thomas et al., 1993). The assessment of ecophysiological characteristics of the tree species might help to determine the degree of risk susceptibility (Vacik et al., 2001). The effects of current environmental conditions on the viability of the population are conducted to describe the risk susceptibility for extinction and the overall viability of the population (see Table 2).

Table 2
Assessment of the current ecological state of the English yew population and the effect of management strategies with regard to the viability of the population

	Relation to evaluation criteria	Effect on viability	Long-term effects of management strategies (planning horizon 20 years)					
			0	I	II	III	IV	V
Environmental characteristics of the present yew population								
Huge number of yew 492 number of species ha^{-1} and the other trees species in total (959 number of species ha^{-1})	Leads to high inter specific competition	−−−	−−−	−−−	+	+	++	++
	Causes moderate intra specific competition Can increase the probability of demographic and genetic change processes and a high genetic sustainability	−	−	−	+	+	++	++
	Assures continuous recruitment by	++	−−	−−	+	+	+++	+++
	continuous production of seeds	+++	+	+	++	++	+++	+++
	Causes a lack of light availability which has negative influences on assimilation, growth and strobilus development	−−−	−−−	−−−	+	+	++	+++
Number of male and female individuals show a female biased sex ratio (1.56)	Which fosters continuous production of seeds as the dioecious sexual system needs a balances ratio of male and female	+++	+	+	+	+	+	+
	Can increase genetic sustainability as the pollination is related to the distribution of male and female individuals	++	0	0	++	++	+++	+++
97.1% of the yew population belong to the third tree layer and 2.9% to the second layer	Which indicates gaps in the vertical structure of the population, yew is missing in the dominant height class	−−	−−	−−	+	+	++	++
A narrow diameter distribution and an average DBH of 8.8 cm	Limits population viability as a broad DBH structure is missing	−−	−−	−−	+	+	++	++
Natural regeneration (>30 cm height < 150 cm) is missing	Which hampers the balanced structure of the population, due to the absence of natural regeneration no supplement trees can build up the future population	−−	−−	−/+	+/−	+/−	+++	+/−
The very vital to vital condition of more than 79% yew trees	Causes continuous production of seeds as good vitality indicates healthy condition	++	0	+	++	++	+++	++
	Reduce the diseases susceptibility	+	0	+	+	+	++	+
	Supports the inter specific competition as it helps to compete with other tree species for growth and development	+	0	+	++	++	+++	++
	Supports the intra specific competition as it increases the overall fitness of the population	−	0	+	++	++	+++	++
Socio-economic characteristics of the present yew population								
Intensive harvest operation in conservation management	Cause soil disturbance which increase the likelihood of juvenile establishment	+	−	−	+	+	++	+++
	Cause damages to the pole stand and increase vulnerability/risk for diseases	−−	+	+	−	−	−	−−−
	Cause damages to the regeneration and increase vulnerability/risk for diseases	−	+	+	−	−	−	−−−
Low people awareness causes direct (illegal cutting) and indirect (browsing by game) human disturbances	Which increases the risk of illegal cutting	−	−	++	−	−	++	−
	Which increases the risk of browsing and leads to an absence of seedlings	−−−	−−−	+++	−−−	−−−	+++	−−−
Regulations according to Gene conservation forests	Increase public acceptability which reduces human pressures (illegal cutting, browsing)	++	−−	++	−−	−−	++	−−
	Causes investments as additional money is needed to maintain the viability of the gene conservation forest and income is reduced by conservation activities	+	+	−−	−	−	−−−	−

(+) Positive effect, increase viability; (−) negative effect, decrease viability; (0) no effect.

2.3. Development of conservation management alternatives

A range of management options should reflect all possible activities for maintaining the species at risk, from minimal protection to a maximum of conservation activities. According to the analysis of the current environmental condition the major threats for the population are identified and management strategies are developed based on the know how, experiences of local experts, and managers (see Vacik et al., 2001; Abiyu et al., 2006). This approach requires the identification of key conservation management decision sequences that affect environments and populations as well as the estimation of various responses of the species to each management decision. The PVRM concept allows the evaluation of the potential effects of management activities on the viability of populations from alternative courses of action.

2.4. Evaluation of viability effects of alternative management

This step entails estimating the likelihood of continued existence for the population in the planning area if the conservation activities are implemented. The effect of management alternatives on the viability of the population with regards to size, distribution, persistence and resources are analysed. Unfortunately such data and models are not available and an extrapolation from existing data sets and qualitative expert knowledge is used to predict the effects on the viability. Different techniques like Decision Trees (e.g. Vacik et al., 2001), Scoring (Given, 1994) or the AHP (Abiyu et al., 2006) can be used.

The AHP provides a framework for selecting an appropriate conservation strategy by means of a decision hierarchy of interrelated evaluation criteria (Mendoza and Prabhu, 2000). The strategies are at the lowest level of the hierarchy and evaluated against all criteria which have an influence on the viability of the population. To evaluate the effects on the viability pair wise comparisons are made between all strategies on a scale of relative importance where the decision maker has the option to express the preferences between two elements on a ratio scale from all elements equal likely (equivalent to a numeric value of 1), to one element being absolute likely (equivalent to a numeric value of 9) compare to others (Saaty, 1995). The overall likelihood of a conservation strategy to maintain an endangered population is determined by synthesizing the individual judgments based on the priorities of the pairwise comparisons and the relative weights of the decision criteria.

2.5. Array and select an alternative

Based on the general evaluation of the conservation strategies to maintain an endangered population the sensitivity analysis allows determination of an overall compromise solution. As uncertainties and assumptions have to be considered in the decision making process the effect of the evaluation criteria on the overall preference of a strategy has to be studied (Reed et al., 2002). The analysis of the effects of different sets of relative weights for the evaluation criteria used in the AHP helps to reduce the uncertainty involved in the process. Finally the array of acceptable alternatives that meet the decision criteria are identified and selected for the course of action.

2.6. Implementation and monitoring

In the implementation phase the actions of the favoured management alternative are implemented by the local forest authority or conservation managers. As a consequence long term monitoring will allow evaluation of whether the assumptions used in PVRM approach have been valid or not.

3. Application

Our present study follows the PVRM concept in order to select an appropriate management strategy for an endangered tree species in Austria.

3.1. Species at risk

In our study the target species is English yew (*Taxus baccata* L.), a slow growing long lived, evergreen, dioecious, wind pollinated-gymnosperm tree which is catalogued as a rare and endangered species, prone to extinction from Austria (Niklfeld, 1999; Schadauer et al., 2003; Russ, 2005) and elsewhere (Svenning and Magård, 1999; Thomas and Polwart, 2003). It is scattered throughout Europe (Bolsinger and Lloyd, 1993), northern Africa (Sauvage, 1941) to the Caspian region of southwest Asia (Mossadegh, 1971). It has gained a considerable importance as a source of anti-cancer drugs (Rikhari et al., 1998) and high aesthetic value (Dovciak, 2002) of timber, but at present declining throughout most of its range (Tittensor, 1980; Hulme, 1996; Dhar et al., 2007). The reasons for decline of yew refer to the over use in the past centuries as well as unsuccessful regeneration, browsing pressure, illegal cutting and lack of appropriate management strategies (Thomas and Polwart, 2003; Dhar et al., 2006). An overview of the ecophysiological characteristics of yew and how they influence the viability of a population are given in Table 1.

3.2. Present status of the yew population in Stiwollgraben

The Austrian gene conservation forest network is used for maintaining the biodiversity of endangered tree species populations in Austria (Müller and Schultze, 1998). Among these gene conservation forests ‘‘Stiwollgraben’’ is located in the southeastern part of Austria about 20 km far from the city of Graz. The primary focus of this forest is the in situ conservation of English yew (*T. baccata* L.) by silvicultural treatments. The assessment of the current population is based on an intensive study of the environmental characteristics of the population (Dhar et al., 2006) which allowed a detailed survey of the population’s structure, growth potentiality, regeneration capa-

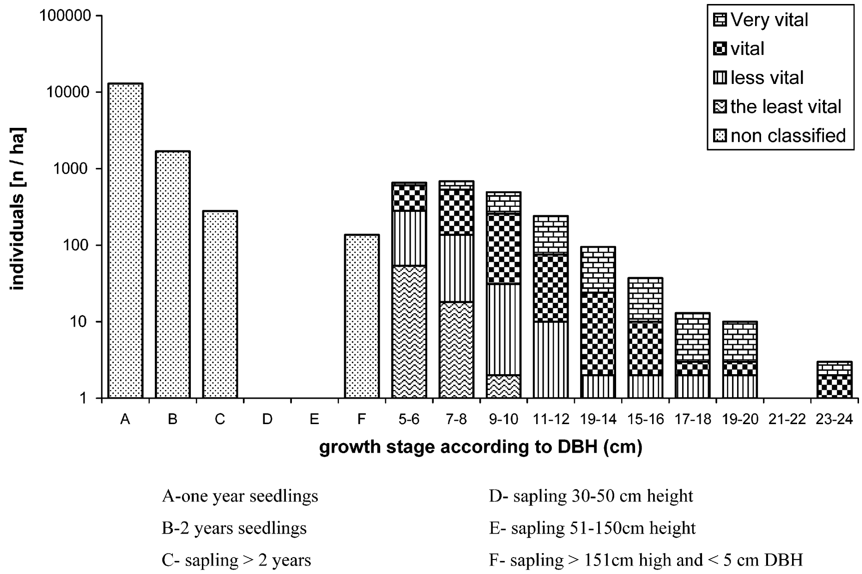


Fig. 1. Vitality of yew population according to the diameter distribution in logarithmic scale (modified from Dhar et al., 2006).

city and negative aspects that influence the viability of the population (e.g. browsing, illegal cutting, competition, biotic and abiotic factors, see Table 2). The forest occupies 4.5 ha of land with a mixture according to basal area of *Fagus sylvatica* (34%), *Picea abies* (23%), *Pinus sylvestris* (15%), *Taxus baccata* (12%), *Larix decidua* (11%) and isolated individuals of *Abies alba*, *Acer pseudoplatanus*, *Ulmus glabra*, *Fraxinus excelsior* and *Sorbus aria* comprising 959 tree individuals ha^{-1} with a total volume of 418 m^3 ha^{-1}. In total 2 236 yews with DBH $\geq$ 5 cm have been found, whereas the vitality condition of the pole stand is very good, more than 79% trees were assessed as very vital to vital whereas 17.4% of the yew belongs to the less vital and 3.4% represent the least vital (see Dhar et al., 2006). The tree height ranges from 1.6 m to 14.4 m with a narrow diameter distribution from $\geq$5 cm to 25 cm. However, saplings in the height class 30–50 cm and 51–150 cm were not observed although 13,019 one-year-old seedlings ha^{-1} were found (Fig. 1). The top height of the dominant tree species was on the average 28 m. As a consequence of the current environmental situation the main problems were interspecific competition with other dominant and co-dominant tree species, illegal cutting by man, lack of natural regeneration due to browsing and probably the loss of genetic variation (Lewandowaski et al., 1995). For details on the environmental characterisation please refer to Dhar et al. (2006).

3.3. Development of management strategies

According to the analysis of the current environmental condition the major threats for the yew population at the gene conservation forest in ‘‘Stiwollgraben’’ have been identified in Table 2. All management strategies (Table 3) are based on the result of personal field experience and different research activities on this tree species in Austria (Vacik et al., 2001; Dhar et al., 2006) as well as throughout the world (Czartoryski, 1978; Mitchell, 1988; Campbell and Nicholson, 1995; Svenning and Magård, 1999; Thomas and Polwart, 2003). The idea was to develop different strategies by altering the main characteristics driving the viability of the population. Therefore we combined various activities in order to develop six management strategies (Table 3). The ‘do nothing’ option (0) keeps the forest stand untouched in order to have a reference. In the ‘wildlife’ option (I), we assumed that building of an exclusion fence for game control and public awareness will be a good combination for conserving the yew population with less human intervention. However strategy no (II) ‘minimum strategy’ deals with the

Table 3
Strategies for conservation management of English yew population

Characteristics	Strategies					
	0 (do nothing)	I (wild life strategy)	II (minimum strategy)	III (single tree selection system)	IV (conservation strategy)	V (timber production strategy)
Wild Life management	No	Fence + game control	No	Fence	Fence + game control	No
Thinning intensity	No	0%	10%	No	30%	50%
Selective thinning	No	No	No	Yes	Yes	No
Careful harvesting	No	No	Yes	Yes	Yes	No
Site preparation	No	No	No	Yes	No	No
Regeneration	Natural	Natural	Natural	Natural	Natural + artificial	Natural + artificial
Public awareness	No	Yes	No	No	Yes	No

decrease of the competition by thinning and applying careful harvesting in order not to hamper the health of the yew trees. The ‘single tree selection’ option (III) is a combination of silviculture measures including selective felling and active soil preparation as well as control of herbivore pressure by building exclusion fence. The ‘conservation’ option (IV) is more related to protection instead of forest production. It covers continuous tree felling and reduction of 30% total stand volume during the management action. Additionally it includes activities like hunting and raising public awareness, which might be help to enhance the acceptability of the local people. The ‘Timber production’ (V) option reduces the competition with dominant and co-dominant species by felling of all other tree species and also includes artificial regeneration for increasing the genetic variation.

3.4. Evaluation of management strategies

3.4.1. Hierarchy structure of forest conservation priority evaluation

The AHP has been used to select an appropriate conservation strategy by means of a decision hierarchy of evaluation criteria and qualitative expert judgements. To identify how the management alternatives affect the viability of the population with regard to size, distribution, persistence, and resources the evaluation hierarchy was developed on the basis of the general environmental analyses of Tables 1 and 2. According to the PVRM concept Table 2 describes the possible causes and effects in relation to risk susceptibility and viability of the yew population. The criteria ‘‘genetic sustainability’’, ‘‘vitality of pole stand’’, ‘‘establishment and viability of seedlings’’ and ‘‘socio-economic factors’’ have been linked to the viability of the population. The indicators (e.g. risk factors, light availability) were selected according to the degree of risk susceptibility of the yew population in relation to general ecological state respectively. The relative weight for each indicator of the parent criterion has been determined by pairwise comparisons of the local foresters considering the actual importance of the factors in the case study area. In that context for the criterion ‘vitality of pole stand’, indicators ‘damage during cutting’ had the highest importance (0.53), with respect to the sub criterion ‘risk factor’. On the other hand ‘spatial structure’ achieved the highest importance (0.57) regarding the sub criterion ‘balanced structure’ (Table 4).

The six management strategies at the lowest level of the hierarchy are evaluated against all indicators, influencing the viability of the population. The effect of each management strategy was qualitatively assessed by an expert team of conservation and forest management specialists (land owner, foresters of the local forest authority, forest experts, scientific researchers) according to each indicator in the context of a workshop in the case study area (Table 2). As a result of this workshop the number of symbols for each indicator indicates the intensity of the effect of the management strategies for increasing (+) or decreasing (−) the viability. These qualitative assessments have been used by the authors as an input for the pairwise comparisons of the management strategies according to each element in the AHP model at the lowest level of the evaluation hierarchy. Pairwise comparisons have been made between all strategies on a scale of relative importance between two strategies on a ratio scale from all elements equal likely (equivalent to a numeric value of 1), to one element being absolute likely (equivalent to a numeric value of 9) compared to others. The overall likelihood of a management strategy to

Table 4
Priorities of criteria and indicator in the general AHP model and for different scenarios

Criteria	Priority	General priorities			
		Sub criteria		Indicator	
Genetic sustainability	0.25	Maintenance of GS	0.50		
		Enhance of GS	0.50		
Vitality of pole stand	0.25	Risk factors	0.32	Damage during cutting	0.53
				Diseases	0.08
				Bark peeling	0.16
				Illegal cutting	0.23
		Competition	0.56	Intraspecific	0.50
				Interspecific	0.50
		Balanced structure	0.12	Vertical structure	0.33
				Spatial structure	0.57
				DBH structure	0.10
Establishment and viability of seedlings	0.25	Light availability	0.34		
		Risk factors	0.30	Browsing	0.68
				Harvesting damage	0.24
				Diseases	0.08
		Soil disturbances	0.06		
		Balanced structure (height)	0.09		
		Continuous production of seeds	0.21		
Socio economic factors	0.25	Know how	0.10		
		Public acceptability	0.30		
		Investment	0.60		

Table 5
Priorities for criteria of scenarios A–E as well as overall priorities, rank and average for management strategies 0–V

Scenario	Priorities of criteria for scenarios				Overall priorities and rank for management strategies					
	Genetic sustainability	Vitality of pole stand	Survival rate of seedlings	Socio economic condition	0	I	II	III	IV	V
A	0.25	0.25	0.25	0.25	5 (0.100)[a]	6 (0.098)	4 (0.114)	3 (0.187)	1 (0.286)	2 (0.214)
B	0.70	0.10	0.10	0.10	5 (0.066)	6 (0.065)	4 (0.088)	3 (0.198)	1 (0.369)	2 (0.215)
C	0.10	0.70	0.10	0.10	6 (0.072)	4 (0.111)	5 (0.111)	2 (0.212)	1 (0.346)	3 (0.148)
D	0.10	0.10	0.70	0.10	6 (0.065)	4 (0.122)	5 (0.097)	2 (0.228)	1 (0.296)	3 (0.191)
E	0.10	0.10	0.10	0.70	2 (0.198)	6 (0.089)	3 (0.161)	5 (0.108)	4 (0.140)	1 (0.303)
Average rank					4.8	5.2	4.2	3.0	1.6	2.2

(A) Overall result, equally prioritized; (B) priority on genetic sustainability; (C) priority on pole stand; (D) priority on survival rate of seedlings; (E) priority on socio economic condition.

[a] Numbers in parentheses indicating the overall priorities for the management strategies according to each scenario.

maintain the endangered population was determined by synthesizing the individual judgments based on the priorities of the pairwise comparisons and the relative weights of the decision criteria and indicators.

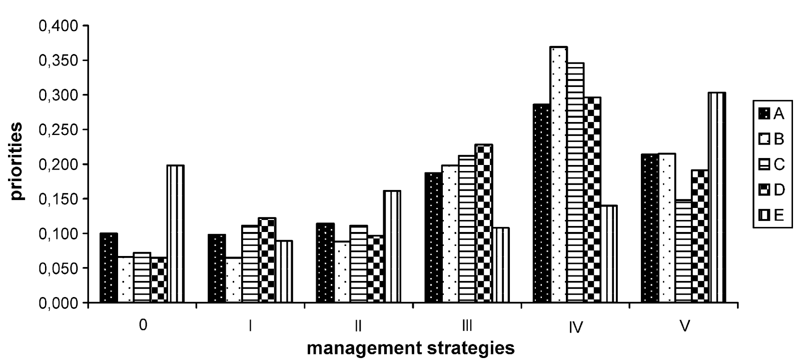
3.5. Selection of management strategies by sensitivity analysis

In order to select a suitable management strategy with increases the viability for the yew population, different environmental scenarios has been evaluated which regard to overall objective. The scenarios are characterized by different combinations of priorities for the evaluation criteria genetic sustainability, vitality of pole stand, establishment and viability of seedlings and socio-economic factors. Scenarios A reflect the actual situation of the criteria influencing the viability of the population. Scenarios B–E indicate different priorities for the evaluation criteria. In each of the scenarios one of the criterion was set to a maximum value to observe the impact of each scenario for the viability of the yew population (Table 5). The overall performance of a management strategy was determined by synthesizing the preferences of the original expert judgments and the relative weights for each criterion of the five scenarios.

In applying the AHP model for all five scenarios it was possible to identify the sensitivity of the model and the management strategies on the viability effects of the yew population. From the scenario analysis it was predicted that management strategy IV achieved almost the highest priorities in all scenarios except for scenario E (Fig. 2). Strategy V is the second best alternative with regards to scenarios A, B, and E, which is followed by strategy III where scenarios C and D was the second best choice. Whereas strategy I attained the lowest priority among the management strategies with respect of scenarios A, B and E. According to ranks and mean ranks of the management strategies based on the results of the sensitivity analysis, it is evident that strategy IV dominates all other strategies independently from the preferences given to the main evaluation criteria genetic sustainability, vitality of pole stand and viability of seedlings (Table 5).

4. Discussion

The results of the PVRM approach indicate that management strategy IV is the best for maintaining the viability of the yew population in the gene conservation forest at Stillwollgraben. The reasons for the estimated positive effects on the



management strategies
0- do nothing
I- wild life strategy
II- minimum strategies
III- single tree selection system
IV- conservation strategy
V- timber production strategy

scenarios
A- over all priority
B- priority on genetic sustainability
C- priority on vitality of pole stand
D- priority on establishment and viability of seedlings
E- priority on socio economic factors

Fig. 2. Overall priorities of management strategies for different scenarios.

viability of the population are the effects on the genetic variation, the enhanced light availability and the reduced browsing pressure by building a fence. Genetic sustainability might be enhanced by the artificial regeneration of yew which increases genetic variation within the species. The thinning operation might lead to a better environmental condition of the mature yews and will increase the seed production additionally (see Vacik et al., 2001). A balanced structure according to vertical structure, patchiness and DBH-distribution provides a good vigour for the pole stand and increases the light availability on the forest floor additionally. According to Iszkulo and Boratynski (2006) insufficient amount of light under the canopy trees can lead to a reduction of the number of yew seedlings even taking into account the shade tolerance of yew. The risk factor browsing is considered as another important parameter for the viability of yew populations. Kelly (1981) and Sarmaja-Korjonen et al. (1991) noted that yew is very susceptible to browsing effects. Herbivore damage is known to increase the mortality risk of individual plants (Watkinson, 1986), induce dormancy in plants (Ehrlén, 1995) and eliminate species from the plant communities (Moolman and Cowling, 1994). Therefore excluding herbivores from the yew population would result in higher rate of adult survival. The management strategy IV also includes also activities to increase the public awareness. This would lead to a reduced number of illegal cuttings and will increase the overall knowledge concerning the conservation management of yew, which is a very important for the overall viability of the population. In that context many authors reported increased public awareness activities are the most effective way for conservation of yew populations (e.g. Campbell and Nicholson, 1995).

The sensitivity analysis indicated that the ‘‘timber production’’ management strategy (V) seems to be best with respect to socio-economic aspects and achieved an average rank of 2.2 considering the different scenarios. This strategy outperforms the other alternatives with respect to the economic return as well as less investment. The primary objective of this management strategy is not to maintain the yew population, however in some cases such a strategy might ensure the viability of yew populations although it is the second best choice.

The PVRM approach allowed the evaluation of different conservation management strategies in order to maintain species at risk and to identify courses of action, which improve the viability of the population. We could demonstrate the benefits in linking this general procedure for the management of endangered tree species to the formal steps of a decision analysis The multi-criteria method AHP allows the use of both qualitative and quantitative information in comparing the importance of criteria or the performance of alternatives. It is possible to consider more qualitative aspects in conservation management like acceptance level knowledge and quantitative information about the productivity and income of different management strategies at the same time. In qualitative assessments of population viability a mix of professional judgment and empirical evidence was used to model population responses to proposed management actions. The effect of environmental and natural changes in certain parts of a population life cycle is a key question both in population management and life history evolution. Therefore it is essential to identify the appropriate instance of a population life cycle where most management efforts should be taken in order to ensure and maintain the maximum population viability. Sensitivity analysis with the AHP is therefore a method of choice when considering uncertain information and risks in conservation management and a large quantity of field data is missing.

It is evident that maintaining the population of yew (*T. baccata*) will require intensive management to prevent its extinction. A more appropriate management programme will have to be implemented to consider the factors influencing the viability of the yew population. In addition, herbivores must be excluded from the population to eliminate the detrimental effect of herbivores damage (Pfab and Witkowski, 2000). Based on the results of the present PVRM approach the local foresters have carried out different treatments in order to maintain the yew population at this site. Different thinning operations, a fence for reducing the browsing pressure and other conservation activities have implemented. With the help of the research a long term monitoring programme was set up, to evaluate the positive effects of the conservation activities on the viability of the population.

Nevertheless, forest ownership is an additional crucial issue in conservation management as in Austria almost 75% forest are privately owned. Therefore it is important that government provides long-term financial incentives for maintaining forest resources in gene conservation forests.

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